
CommunityKV: Efficient Long-Context Decoding via Graph Partitioning

Anonymous Author(s)

Affiliation

Address

email

Abstract

Scaling Transformers to long contexts is constrained by the quadratic cost of self-attention and the linear growth of key-value cache memory transfer. Sparse attention mitigates this by retrieving only relevant tokens, but current approaches either require large-scale training or, within the training-free regime, rely on semantically coarse heuristics or expensive clustering that is difficult to update efficiently during decoding. We introduce CommunityKV, a framework that formulates sparse attention as a community detection problem. CommunityKV constructs a token graph from the QK^T scores already computed during standard prefill, and partitions the graph into communities to enable retrieval of semantically coherent token groups. A local update rule assigns newly generated tokens to communities in constant time, enabling sparse retrieval throughout streaming decoding without global re-partitioning. We evaluate CommunityKV on open weight models and long context benchmarks, demonstrating that our approach achieves up to $1.86\times$ higher decoding throughput than exact FlashAttention-2 with negligible accuracy degradation.¹

1 Introduction

The capacity for Transformers [36] to process long context inputs powers a new generation of applications, ranging from repository-level code understanding [2, 29] and long-horizon agentic planning [11] to multimodal reasoning [38]. However, as the context length grows, standard self-attention encounters prohibitive bottlenecks: the quadratic cost of attention and the linear growth of key-value (KV) cache memory transfer.

Despite the increasing KV cache size for modern applications, empirical studies demonstrate that token prediction accuracy is driven by only a small fraction of relevant past tokens [46, 14]. Sparse attention methods attempt to exploit this redundancy by loading only a portion of the KV cache at each generation step [34, 26]. Recent architectural methods such as DeepSeek Sparse Attention [9] integrate sparsity into the training objective, but require large-scale training. Within the training-free regime, current approaches force a compromise: they either employ retrieval heuristics that suffer from unacceptable accuracy degradation [33] or rely on heavy preprocessing of the KV cache that is difficult to update during decoding [16].

In this work, we propose CommunityKV, a framework that formulates training-free sparse attention as community detection on a graph induced from attention scores. CommunityKV leverages the structure already present in the QK^T matrix to identify semantically coherent token groups. A local update rule extends the partition to newly generated tokens in constant time, enabling streaming decoding without re-clustering.

¹Code is available at https://anonymous.4open.science/r/community_kv-neurips_2026.

35 Our contributions are as follows:

- 36 • **Formulation:** We formulate sparse attention as community detection on a token graph
37 induced from $QK^\top$ scores. A combined graph captures both direct query-key dependencies
38 and second-order co-attention structure, enabling retrieval of semantically coherent token
39 groups without auxiliary embeddings or vector-space clustering. A local update rule extends
40 the partition to newly generated tokens, supporting streaming decoding without global
41 re-partitioning.
- 42 • **Complexity analysis:** We show that graph construction is amortized within the standard
43 prefill matrix multiplication, and partitioning adds only $\mathcal{O}(n \log n + nd)$. During decoding,
44 community retrieval is sub-linear at $\mathcal{O}(\sqrt{n}d)$ per step, and adding a new token to the
45 partition is $\mathcal{O}(d)$, independent of sequence length, achieving sub-linear retrieval with
46 constant-time updates simultaneously.
- 47 • **Empirical results:** We evaluate on long context benchmarks with open weight models of
48 varying sizes, demonstrating that CommunityKV matches dense FlashAttention-2 accuracy
49 while delivering up to $1.86\times$ decoding speedup at less than 8% prefill overhead. Accuracy
50 improves with retrieval budget, outperforming both graph-based KV cache compression and
51 existing training-free retrieval-based sparse attention methods.

52 2 Related work

53 **Architectural vs. training-free sparse attention** Recent methods integrate sparsity directly into
54 the training objective, achieving large throughput gains. DeepSeek Sparse Attention (DSA) [9] scores
55 queries against lightweight indexer keys to select a top-k subset of past tokens for full attention,
56 while Native Sparse Attention (NSA) [43] combines block-level token compression with fine-grained
57 token selection. However, these approaches couple sparsity to model training: the upfront cost
58 amortizes only at very high inference volumes, and the resulting sparsity pattern is fixed at training
59 time, limiting retrieval quality to the training distribution.

60 CommunityKV targets the complementary setting: training-free sparse attention. Within the training-
61 free regime, existing approaches reduce KV cache traffic via two strategies. KV cache compression
62 methods [46, 22, 24] enforce strict memory ceilings through token eviction or merging, but are
63 inherently lossy: once discarded, information cannot be recovered. Selective retrieval methods
64 instead preserve the full KV cache in memory and dynamically load only relevant tokens at each
65 decoding step. The effectiveness of retrieval-based methods depends on the design of the sparsity
66 mask, which typically employs either fixed or dynamic patterns.

67 **Fixed-pattern sparse attention** Early work on sparse attention relied on predefined, query-
68 independent sparsity patterns. The Sparse Transformer [7] introduced factorized attention with
69 strided patterns, achieving sub-quadratic $\mathcal{O}(n\sqrt{n}d)$ complexity. Longformer [4] and BigBird [44]
70 combined sliding windows, dilated patterns, and global tokens to achieve linear $\mathcal{O}(nd)$ complexity.
71 LongNet [10] extended dilated attention to billion-token sequences, and StreamingLLM [41] showed
72 that retaining initial “attention sinks” stabilizes sliding window attention. Because these patterns are
73 fixed in advance, they miss relevant tokens that fall outside the predefined structure.

74 **Dynamic sparse attention** Dynamic sparse attention frames the problem as query-dependent
75 retrieval, typically using auxiliary data structures to index the KV cache and select relevant subsets at
76 each step. A first family uses heuristics over contiguous token blocks. Quest [33] tracks channel-wise
77 min-max values to estimate query-key overlap, while a similar training-based method MoBA [25]
78 applies block-wise gating to filter context chunks. These methods avoid heavy auxiliary data structures
79 but treat blocks of contiguous tokens as the retrieval unit: a high-scoring outlier within a block triggers
80 retrieval of the entire block, including unrelated neighbors. A second family uses Locality Sensitive
81 Hashing [17, 45, 6, 15] to bucket tokens using random projections. Because these projections are
82 data-independent, they fail to account for the distribution of key embeddings; achieving high recall
83 requires scaling the number of hash tables, which re-introduces significant overhead. A third family
84 applies k -means clustering to the key or value vectors [30, 16, 37, 39]. These methods produce
85 semantically coherent clusters, but k must be chosen a priori and centroids drift as new tokens are
86 generated, forcing a trade-off between stale clusters and expensive re-clustering.

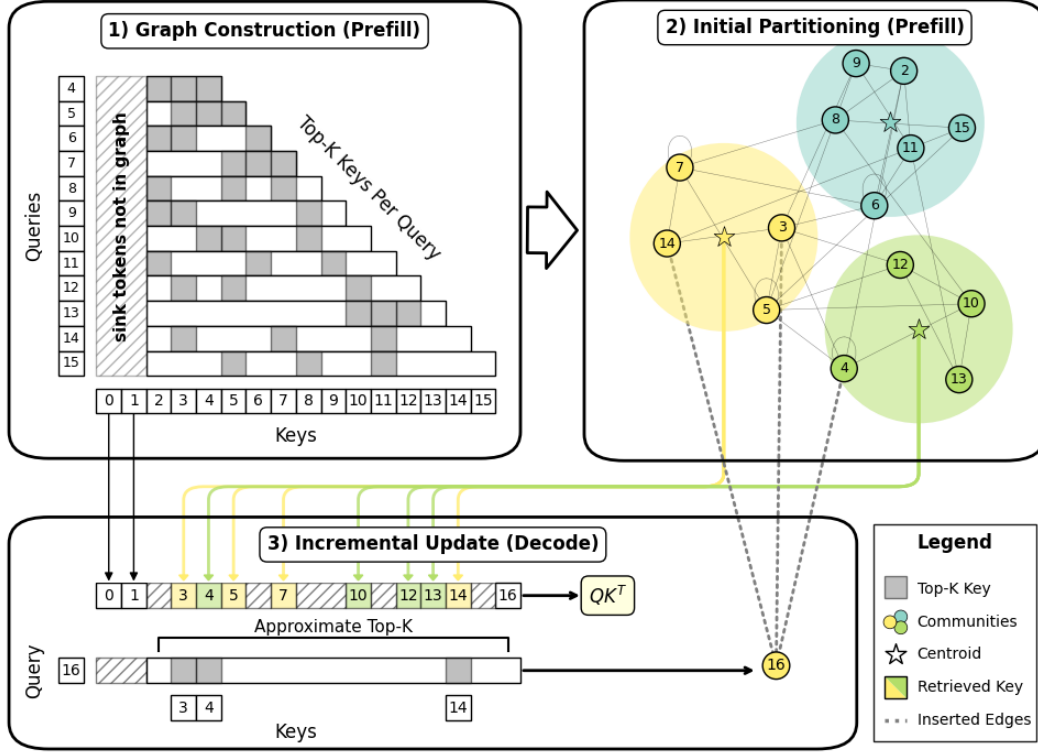


Figure 1: Overview of the CommunityKV framework. **(1) Graph Construction.** During prefill, we induce a sparse graph from the attention matrix by connecting each query to its top- κ keys (gray cells, $\kappa = 3$ shown). Sink tokens (hatched) are excluded from the top- κ pool. **(2) Initial Partitioning.** We partition the graph into semantic communities (colored regions) using the Leiden algorithm and compute a centroid (star) per community. **(3) Incremental Update.** At each decoding step (token 16 shown), we score the query against community centroids to retrieve a candidate subset, run exact attention on this subset together with sink tokens, and use the resulting top- κ keys to update the new token’s edges and assign it to a community in constant time. For clarity, only direct attention $w^{(1)}$ relations are depicted; this corresponds to $\lambda = 1$ in the combined graph.

87 **Comparison to GraphKV** GraphKV [21] is the most closely related work, constructing a sparse
88 attention graph and applying a “decay signal propagation” algorithm to evict semantically redundant
89 tokens. The result is a compressed cache that every subsequent decoding step queries against.
90 CommunityKV adopts the opposite philosophy. Instead of treating semantic similarity as a signal
91 to be decayed, we treat it as the basis for community formation: we retain the full KV cache and
92 dynamically retrieve communities for each query, allowing the model to access locally dense semantic
93 structure without permanently deleting context.

94 3 Method

95 Our approach has three phases (Figure 1): (1) graph construction, where we build a sparse token
96 graph from attention scores during prefill; (2) initial partitioning, where we cluster the graph into
97 semantic communities; and (3) incremental update, where newly generated tokens are assigned to
98 communities in constant time during decoding.

99 3.1 Graph construction

100 We construct a graph where nodes represent tokens and edges encode two kinds of relations: direct
101 attention, where one token attends strongly to another, and co-attention, where two tokens are jointly
102 attended by a common query. Let $\alpha \in [0, 1]^{n \times n}$ be the attention score matrix over an n -token context,

103 where α_{ij} is the score of query i attending to key j , and let $\mathcal{K}_i$ be the top- κ key positions for query i :

$$\mathcal{K}_i = \arg \max_{j=|\mathcal{S}|+1, \dots, i} \alpha_{ij}. \quad (1)$$

104 We exclude the set of initial “sink token” positions $\mathcal{S}$, as they exhibit disproportionately high attention
105 scores regardless of semantic relevance [40].

106 **Direct attention graph** The direct attention graph captures explicit query-key dependencies: a
107 directed edge $i \rightarrow j$ records that query i attends strongly to key j :

$$w_{ij}^{(1)} = \alpha_{ij} \mathbf{1}(j \in \mathcal{K}_i) \quad (2)$$

108 where $\mathbf{1}(\cdot)$ is the indicator function.

109 **Co-attention graph** If two keys are strongly attended by the same query, they play similar roles in
110 that query’s context. The co-attention graph captures this by linking keys that co-occur in the top- κ
111 set of a common query:

$$w_{ij}^{(2)} = \sum_{m=|\mathcal{S}|+1}^n \alpha_{mi} \alpha_{mj} \mathbf{1}(\{i, j\} \subseteq \mathcal{K}_m). \quad (3)$$

112 **Combined graph** We combine the two adjacency matrices into a single symmetric adjacency
113 matrix:

$$w = \frac{\lambda}{2} [w^{(1)} + (w^{(1)})^\top] + (1 - \lambda) w^{(2)}, \quad (4)$$

114 with $0 \leq \lambda \leq 1$ controlling the balance between direct and co-attention relations. By construction,
115 the graph is sparse: each of the n queries contributes κ edges to $w^{(1)}$ and at most κ^2 edges to $w^{(2)}$,
116 so the total edge count satisfies $E = \mathcal{O}(n\kappa(1 + \kappa)) = \mathcal{O}(n)$ for constant κ .

117 **Theoretical justification** The combined graph w admits a dynamical interpretation that motivates
118 graph partitioning, grounded in the mathematical structure of self-attention. Because α is row-
119 stochastic, it defines the one-step transition kernel of a discrete Markov chain over the token sequence.
120 Symmetrizing as $(\alpha + \alpha^\top)/2$ and combining with the second-order affinity matrix $\alpha^\top \alpha$ yields a
121 symmetric kernel of a time-reversible random walk, and sparsifying via top- κ selection produces w ,
122 preserving the dominant transition paths with bounded sparsity ($E = \mathcal{O}(n)$).

123 In this framework, token communities correspond to metastable states of the walk: groups of
124 tokens the walk is unlikely to escape over time. We partition the graph into these communities by
125 maximizing modularity [27], which on the symmetric graph defined by w is equivalent to maximizing
126 the linearized Markov Stability of the induced walk [20]. In practice, we employ the Leiden
127 algorithm [35] to perform this graph partitioning.

128 3.2 Initial partitioning

129 **Leiden community detection** The Leiden algorithm [35] iterates three phases: fast local moving,
130 refinement, and aggregation, to maximize modularity:

$$Q = \frac{1}{2m} \sum_{ij} \left(w_{ij} - \gamma \frac{w_i w_j}{2m} \right) \mathbf{1}(c_i = c_j), \quad (5)$$

131 where $m = \frac{1}{2} \sum_{ij} w_{ij}$ is the total graph weight, $w_i = \sum_j w_{ij}$ is the weighted degree of token i , and
132 c_i is the community label of token i . The resolution parameter γ controls community granularity:
133 larger values favor smaller, denser communities. Leiden improves upon Louvain [5] by guaranteeing
134 connected communities.

135 We choose Leiden over alternative community detection algorithms for several reasons. Leiden
136 empirically exhibits $\mathcal{O}(E)$ runtime [32, 35] and few iterations required for convergence; [5] report
137 fewer than 5 across all tested networks for the analogous Louvain algorithm. We cap iterations at
138 $r = \lfloor \log_{10} n \rfloor$, yielding $r \in [4, 7]$ for our evaluation contexts (10k–10M tokens). With graph sparsity
139 bounded at $E = \mathcal{O}(n)$, graph partitioning costs $\mathcal{O}(rE) = \mathcal{O}(n \log n)$. We then compute community

Table 1: Computational complexity comparison. n is sequence length, d is hidden dimension. Note that our method achieves sub-linear retrieval $\mathcal{O}(\sqrt{n}d)$ without expensive clustering or update costs.

Method	Initial Clustering	Per-step Update	Per-Step Retrieval
Quest	$\mathcal{O}(nd)$	$\mathcal{O}(d)$	$\mathcal{O}(nd)$
Multipole	$\mathcal{O}(n^2d)$	$\mathcal{O}(nd)$	$\mathcal{O}(nd)$
CommunityKV	$\mathcal{O}(n \log n + nd)$	$\mathcal{O}(d)$	$\mathcal{O}(\sqrt{n}d)$

centroids in a single $\mathcal{O}(nd)$ pass, yielding a clustering cost of $\mathcal{O}(n \log n + nd)$ (Table 1). By contrast, Multipole Attention assigns n tokens to $\mathcal{O}(n)$ centroids via k -means, yielding $\mathcal{O}(n^2d)$ clustering cost. Because pairwise similarity is already captured in the graph edges via $QK^\top$, our clustering operates on scalar weights rather than d -dimensional vectors.

Leiden’s local-moving phase is embarrassingly parallel; each node’s modularity-gain computation ΔQ (Eq. 7) depends only on its neighbors and community-level aggregates, so all nodes can be evaluated simultaneously. Recent parallel implementations exploit this structure to achieve a processing rate of 403M edges/s on a 3.8B-edge graph using 64 CPU threads [32], and NVIDIA’s cuGraph [12] reports significant speedup over CPU baselines.

Leiden admits incremental update schemes that avoid global re-partitioning when edges are added [31, 23]. Our local update rule (Section 3.3) is a domain-specific specialization of this idea, exploiting the structure of autoregressive decoding to assign new tokens in $\mathcal{O}(d)$ time per step.

3.3 Incremental update

Community retrieval During decoding, we retrieve relevant tokens by scoring the latest query q_i against community centroids and aggregating positions from the top-ranking communities into a candidate set $\mathcal{R}_i$. We then form the sparse context $\mathcal{S} \cup \mathcal{R}_i \cup \{i\}$ by augmenting $\mathcal{R}_i$ with the attention sinks $\mathcal{S}$ and the current query position i , truncating to a fixed token budget. Because our graph satisfies $E = \mathcal{O}(n)$, the modularity resolution limit [13] implies the number of communities scales as $\mathcal{O}(\sqrt{n})$ (verified empirically in Figure 2), making retrieval cost $\mathcal{O}(\sqrt{n}d)$ (Table 1).

Edge approximation To integrate a new token i into the graph, we approximate its edge weights using only the retrieved set. We compute approximate attention scores and top- κ keys over $\mathcal{R}_i \cup \{i\}$:

$$\begin{aligned}\tilde{\alpha}_{ij} &= \text{softmax}(q_i^\top k_j)_{j \in \mathcal{R}_i \cup \{i\}}, \\ \tilde{\mathcal{K}}_i &= \arg \text{top-}\kappa \tilde{\alpha}_{ij}. \end{aligned} \quad (6)$$

Substituting these approximations into Eqs. (2) and (3) yields the edge updates. We add κ direct attention edges $w_{ij}^{(1)} = \tilde{\alpha}_{ij}$ for each $j \in \tilde{\mathcal{K}}_i$, and increment the κ^2 co-attention edges between pairs $j, k \in \tilde{\mathcal{K}}_i$ by $\Delta w_{jk}^{(2)} = \tilde{\alpha}_{ij} \tilde{\alpha}_{ik}$.

Greedy community assignment After integrating the new token i into the graph, we assign it a community label. We freeze the existing community assignments and consider two options: form a new singleton community $\{i\}$ or join a neighboring community $C \in \{c_j \mid j \in \tilde{\mathcal{K}}_i\}$. We pick the option that maximizes the modularity gain:

$$\Delta Q(i \rightarrow C) \propto \sum_{j \in C \cap \tilde{\mathcal{K}}_i} w_{ij} - \gamma \frac{w_i w_C}{2m}, \quad (7)$$

where $w_C = \sum_{j \in C} w_j$ is the total weight of community C .

Each decoding step touches at most $\kappa(\kappa + 1)$ edges, evaluates κ candidate communities, and updates one centroid in $\mathcal{O}(d)$. Since κ is a small constant, the total per-step update cost is $\mathcal{O}(d)$, independent of sequence length n (Table 1).

Stability of incremental updates Our local update rule maintains partition quality throughout decoding without global re-partitioning. In a stress test over 4096 generated tokens, modularity

remains stable while attention mass decayed gracefully staying above 85% at a 4096-token budget (Figure 3; see Appendix B for the full experimental protocol). The KV cache and the routing graph are decoupled in CommunityKV: updates modify only the graph, leaving past KVs untouched. A single global re-partitioning restores optimal modularity in the graph. Running Leiden every $\mathcal{O}(\sqrt{n})$ steps adds amortized $\mathcal{O}(\sqrt{n} \log n)$ cost per step, making periodic refresh feasible for unbounded generation.

3.4 Implementation

We fuse top- κ selection into the FlashAttention kernel: as each $b_q \times b_k$ tile is computed, a register-level bitonic sort identifies b_q sets of local top- κ candidates, which are reduced per-query via radix select. The total cost is $\mathcal{O}(n^2)$, matching the underlying prefill matrix multiplication, with constant factors determined by the tile dimensions b_q , b_k and κ (Appendix C.1). Empirically, this adds 6–8% to prefill wall time (Table 5).

Leiden partitioning is pipelined with the forward pass: as soon as layer L ’s attention scores are available, its graphs are dispatched to asynchronous workers while layers $L+1, \dots$ continue executing (Figure 4). The only exposed latency is the final layer’s partition (0.37–0.45s across the Qwen3 family). We provide a graph construction complexity derivation, pipelining schedule, and memory accounting in Appendix C.

4 Experiments

We evaluate CommunityKV along three axes: (1) accuracy compared to sparse attention baselines on long context benchmarks, (2) wall-clock latency on an NVIDIA H100 80GB, and (3) ablation studies over key design parameters. Unless otherwise noted, we fix $\kappa = 8$, $\lambda = 0.5$, exclude the first 10 sink tokens from graph construction, and maintain one independent graph per query head. All models use greedy decoding (single run, deterministic) with a maximum of 128 generated tokens and YaRN RoPE scaling [28] to extend the model context. To ensure a fair comparison, we tune the resolution parameter γ so that the number of communities matches a baseline cluster count of 16 (see Appendix A for the γ -community-size mapping).

4.1 Experimental setup

4.1.1 Datasets

LongBench v2 We evaluate on LongBench v2 [1], a suite of challenging multiple-choice questions spanning code understanding, long-dialogue history, and multi-document QA.

BABILong We additionally evaluate on BABILong [19] QA1–QA5, a multi-hop reasoning benchmark requiring retrieval of 1–5 supporting facts from long context inputs.

4.1.2 Models

Qwen3 Model Collection We conduct our evaluations using the Qwen3 family [42], specifically the 4B, 8B, and 14B parameter variants.

4.1.3 Baselines

FlashAttention-2 (Dense) Exact FlashAttention-2 [8] serves as the lossless reference for both accuracy and wall-clock latency.

GraphKV As the most closely related graph-based method, GraphKV [21] constructs a sparse attention graph and permanently evicts low-scoring tokens during prefill. We include it to isolate the benefit of dynamic retrieval over static eviction.

We additionally compare against two training-free retrieval methods that span the efficiency-fidelity spectrum:

Quest Quest [33] partitions consecutive tokens into fixed blocks and tracks channel-wise min-max statistics for retrieval, representing the fast but semantically coarse end of the spectrum.

Table 2: Accuracy on LongBench v2 across token budgets (4k–32k) for GraphKV and CommunityKV. Dynamic community retrieval matches dense accuracy at $8\times$ lower budget than static eviction.

<i>Qwen3-8B</i> (Dense: 31.4)				
	4096	8192	16384	32768
GraphKV	3.4	10.5	18.5	22.5
CommunityKV	31.1	31.2	31.4	31.4

Table 3: Accuracy on LongBench v2 across token budgets (128–4096) at cluster size 16 for Qwen3 models (4B, 8B, 14B). CommunityKV accuracy improves monotonically with budget and reaches dense parity at 4096 tokens across all model scales.

	128	256	512	1024	2048	4096
<i>Qwen3-4B</i> (Dense: 27.4)						
Quest	0.2	0.8	3.0	6.6	12.5	14.1
Multipole	25.8	26.2	25.6	25.8	25.6	25.8
CommunityKV	25.4	26.1	25.8	26.6	26.2	27.4
<i>Qwen3-8B</i> (Dense: 31.4)						
Quest	5.6	9.1	15.1	17.5	22.5	24.7
Multipole	29.4	30.0	28.4	29.8	28.2	28.8
CommunityKV	28.5	28.8	29.5	30.1	29.9	31.1
<i>Qwen3-14B</i> (Dense: 32.0)						
Quest	9.3	18.1	21.5	24.7	29.0	30.6
Multipole	30.4	29.2	29.2	28.0	28.8	28.8
CommunityKV	30.2	30.0	30.5	30.7	29.7	32.0

Multipole Attention Multipole [16] clusters keys via k -means, representing the high-fidelity end. It serves as a performance target that our lighter approach seeks to match without the extra compute cost.

4.2 Long context understanding

Graph-based retrieval vs. KV eviction Both CommunityKV and GraphKV construct sparse graphs from attention scores, but differ in how they use the graph: GraphKV evicts tokens permanently, while CommunityKV retrieves dynamically. To isolate this distinction, we compare the two on LongBench v2 with Qwen3-8B, sweeping GraphKV’s budget from 4k to 32k tokens (Table 2). GraphKV requires an $8\times$ larger budget (32k tokens) to reach 22.5% accuracy, while CommunityKV achieves 31.1% with only 4k tokens, matching the dense baseline. This confirms that semantic structure should guide dynamic retrieval rather than permanent context deletion.

Comparison between training-free retrieval methods We compare CommunityKV against Quest and Multipole on LongBench v2 across the Qwen3 family (4B, 8B, 14B), sweeping token budgets from 128 to 4096 at cluster size 16 (Table 3). CommunityKV achieves parity with exact FlashAttention-2 across all model scales at a 4096-token budget; for example, 32.0% on Qwen3-14B, matching the dense upper bound exactly. Multipole performs strongly at restricted budgets (128–256 tokens) but plateaus as the budget increases, suggesting that k -means fails to capture the semantic structure required for complex reasoning. Quest scales with budget but trails dense by over 6% even at 4k tokens. CommunityKV bridges this gap: its accuracy rises steadily with budget, translating additional retrieval capacity directly into better accuracy.

Multi-hop retrieval To test whether community structure aids multi-hop retrieval, we evaluate on BABILong QA1–QA5 with Qwen3-8B at 64k context, using a 4096-token budget for all retrieval-based methods (Table 4). CommunityKV achieves the highest average accuracy among sparse

Table 4: Accuracy on BABILong QA1–QA5 (Qwen3-8B, 64k context, 4096-token budget). CommunityKV achieves the highest average accuracy among sparse methods.

	QA1	QA2	QA3	QA4	QA5	Avg
Dense	0.66	0.38	0.23	0.55	0.62	0.488
Quest	0.38	0.35	0.22	0.49	0.59	0.406
Multipole	0.40	0.26	0.14	0.53	0.60	0.406
CommunityKV	0.50	0.38	0.25	0.54	0.63	0.440

Table 5: Latency comparison between FlashAttention-2 (Dense) and CommunityKV on an NVIDIA H100 80GB at 131k context. Left: total prefill wall time with breakdown into fused top- κ selection and exposed Leiden partitioning latency. Right: average per-step decoding wall time (128 generated tokens) with breakdown into token retrieval and sparse attention. Only attention-related kernels are reported; the remaining per-step time is MLP and layer normalization.

Prefill – Total Wall Time (s)				Decoding – Avg Per-Step Wall Time (ms)			
	4B	8B	14B		4B	8B	14B
Dense	16.5	18.2	26.3	Dense	63	64	71
CommunityKV	17.8	19.6	28.2	CommunityKV	37	34	51
Top- κ Sel.	0.94	1.09	1.49	Retrieval	5.3	4.7	7.2
Leiden	0.37	0.45	0.42	Attention	6.1	5.5	8.2
<i>Overhead</i>	+7.8%	+7.7%	+7.2%	<i>Speedup</i>	1.69×	1.86×	1.39×

242 methods and matches or exceeds the dense baseline on QA2, QA3, and QA5: tasks requiring chaining
 243 two or more supporting facts.

244 4.3 Computational efficiency

245 To measure the latency trade-off, we benchmark prefill and per-step decoding time for Communi-
 246 tyKV against dense FlashAttention-2 on the Qwen3 family (4B, 8B, 14B) at the maximum native
 247 131k context length of Qwen3 models (Table 5).

248 CommunityKV achieves up to $1.86\times$ decoding speedup (8B model) at less than 8% prefill overhead.
 249 The overhead is dominated by fused top- κ selection (0.9–1.5s), most of which is masked by the IO
 250 cost of the 130k-token prefill; the exposed Leiden partitioning adds only 0.37–0.45s. On the decode
 251 side, CommunityKV per-step latency is governed by the fixed retrieval budget rather than sequence
 252 length, yielding near-constant cost throughout the 130k-token window.

253 4.4 Ablations

254 We ablate the key design parameters of CommunityKV on LongBench v2 with Qwen3-14B at a
 255 4096-token budget, starting from the default configuration ($\kappa = 8$, $\lambda = 0.5$, graph per query head, 10
 256 sink tokens excluded) (Table 6).

257 **Graph sparsity (κ)** Accuracy peaks at $\kappa = 8$ (32.0%) and drops sharply at higher values (27.8%
 258 at $\kappa = 32$), indicating that excessive connectivity obscures community boundaries. Too few edges
 259 ($\kappa = 2$, 30.3%) fragments the graph.

260 **Direct vs. co-attention (λ)** Accuracy peaks at $\lambda = 0.5$ (32.0%) and degrades toward pure direct
 261 attention ($\lambda = 1$, 28.5%) or pure co-attention ($\lambda = 0$, 28.8%), confirming that neither signal is
 262 sufficient alone.

263 **Centroid budget** By default, retrieval greedily fills the token budget with tokens from top-ranked
 264 communities. We test reserving part of the budget for additional community centroids as compressed
 265 summaries ($b_c \in \{0, 4, 16, 64, 256\}$). The greedy policy ($b_c = 0$, 32.0%) consistently outperforms

Table 6: Ablation studies on LongBench v2 (Qwen3-14B, token budget=4096). The default configuration ($\kappa=8$, $\lambda=0.5$, per-head graphs, sinks excluded) is marked with gray shading and achieves the best accuracy in every category.

Graph Sparsity (κ)	$\kappa = 2$ 30.3	$\kappa = 4$ 29.4	$\kappa = 8$ 32.0	$\kappa = 16$ 28.7	$\kappa = 32$ 27.8
Graph Weight (λ)	$\lambda = 0.0$ 28.8	$\lambda = 0.25$ 29.0	$\lambda = 0.50$ 32.0	$\lambda = 0.75$ 29.0	$\lambda = 1.0$ 28.5
Centroid Budget	0 32.0	4 27.5	16 29.7	64 28.4	256 28.6
Graph Aggregation	No Agg. 32.0	Query Group 31.8	Layer-Wise 21.5	– –	– –
Sink Tokens	Excluded 32.0	In-Graph 31.3	– –	– –	– –

centroid summaries (e.g., 28.6% at $b_c = 256$): full-token depth is more valuable than compressed breadth.

Graph aggregation In group query attention (GQA) architectures, query heads share KV heads. We test three granularities for graph construction: per-query-head (default), query-group (one graph per KV group), and layer-wise (one graph per layer). Per-head (32.0%) and query-group (31.8%) perform comparably, while layer-wise collapses to 21.5%. Query-group aggregation is a viable memory-saving alternative.

Sink tokens Excluding sink tokens from the graph and always retrieving them (32.0%) outperforms including them in the partition (31.3%). Sinks attract attention from nearly all positions, creating high-degree hubs that bridge unrelated communities.

5 Conclusion

We presented CommunityKV, a training-free framework that formulates sparse attention as dynamic community detection on a token graph induced from $QK^\top$ scores. Partitioning this graph with the Leiden algorithm during prefill yields communities that serve as the retrieval unit during decoding, with sub-linear $\mathcal{O}(\sqrt{n}d)$ cost per step. A constant-time local update rule assigns generated tokens to communities without global re-partitioning, enabling streaming decoding at fixed cost per step.

Empirically, CommunityKV achieves up to $1.86\times$ decoding speedup over FlashAttention-2 at less than 8% prefill overhead, while matching dense accuracy on LongBench v2 and BABILong across the Qwen3 family (4B–14B). It outperforms Quest’s heuristic retrieval and matches Multipole’s k -means clustering at a fraction of the cost.

Limitations and future work Our greedy community assignment freezes prior assignments, which may cause partition drift over very long generation horizons; periodically re-partitioning every $\sqrt{n}$ steps mitigates this at amortized $\mathcal{O}(\sqrt{n} \log n)$ cost per step, but adds latency in practice. The edge approximation during decoding is restricted to the retrieved set, so true top- κ keys outside retrieved communities are missed, and retrieval recall affects the quality of incremental updates. These limitations suggest several natural extensions of the framework.

First, incremental Leiden refinement [23], performing local moves in the neighborhood of newly inserted tokens rather than single-node greedy assignment, could better maintain partition quality during decoding over longer horizons. Second, the hierarchical structure that Leiden naturally produces via its aggregation phase could enable $\mathcal{O}(\log nd)$ retrieval at longer contexts where even $\mathcal{O}(\sqrt{n})$ communities become expensive to score. Finally, sharing community structure across adjacent layers, whose attention patterns are often correlated, could reduce both graph memory and pipelining depth.

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A Community scaling and resolution parameter

We empirically verify two properties of the graph partition on LongBench v2 using Qwen3-14B (Figure 2).

A.1 Resolution parameter γ

The left panel shows the relationship between the Leiden resolution parameter γ and the average community size. Increasing γ produces smaller, denser communities, providing a direct control over retrieval granularity. In our experiments, we tune γ to match a cluster size of 16 for fair comparison with each baseline (Section 4.2).

A.2 Sub-linear community scaling

The right panel shows the number of communities as a function of sequence length. The empirical scaling closely follows $\mathcal{O}(\sqrt{n})$, consistent with the resolution limit of modularity optimization [13]. At 100k tokens, the graph partitions into approximately 50 communities. This sub-linear growth is what enables $\mathcal{O}(\sqrt{n}d)$ retrieval cost: scoring a query against $\mathcal{O}(\sqrt{n})$ centroids is substantially cheaper than the $\mathcal{O}(nd)$ cost of scoring against all tokens.

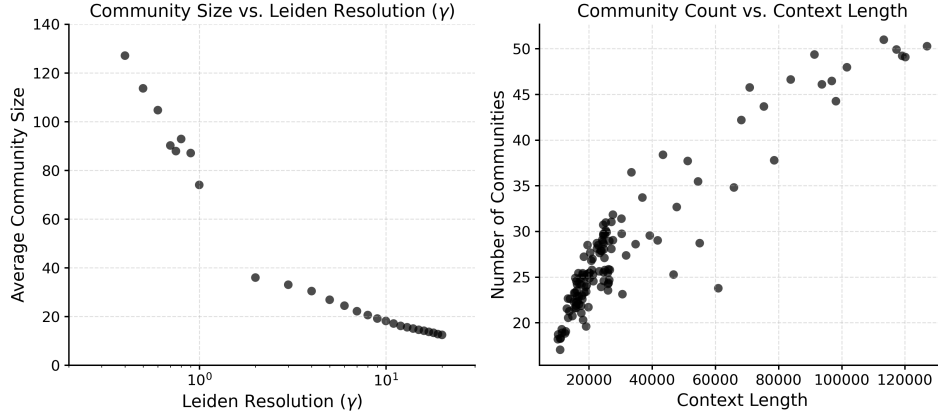


Figure 2: Empirical analysis of graph partition properties on LongBench v2 (Qwen3-14B). **Left:** Average community size as a function of the resolution parameter γ . **Right:** Number of communities vs. context length, confirming $\mathcal{O}(\sqrt{n})$ sub-linear scaling.

B Partition drift analysis

We stress-test the stability of our incremental update rule by tracking two metrics over a 4096-token generation horizon on LongBench v2 with Qwen3-14B, using the default configuration ($\kappa = 8$, $\lambda = 0.5$, 4096-token retrieval budget). No global re-partitioning is performed during generation.

B.1 Metrics

We monitor (1) graph modularity Q (Eq. 5), which measures how well the current partition reflects the evolving graph structure, and (2) attention mass recall $\tilde{\mu}_\kappa = \sum_{j \in \mathcal{R}_i} \alpha_{ij}$, the fraction of the full attention distribution captured by the retrieved set at each decoding step.

B.2 Results

Figure 3 shows both metrics as a function of generated tokens. Modularity remains in $[0.44, 0.50]$ throughout the 4096-token horizon, indicating that the greedy assignment rule continues to improve partition quality as new tokens reinforce existing community structure. Attention mass recall decays gracefully: at the 4096-token retrieval budget, recall remains above 85% after 4096 generated

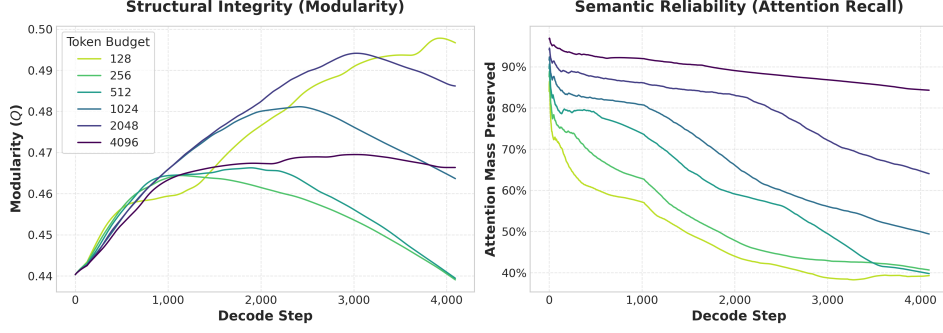


Figure 3: Partition stability over a 4096-token generation horizon (Qwen3-14B, LongBench v2). **Left:** Graph modularity Q remains stable and increases slightly as new tokens reinforce community structure. **Right:** Attention mass recall $\tilde{\mu}_\kappa$ at varying retrieval budgets. At the default 4096-token budget, recall stays above 85% throughout generation.

454 tokens. Smaller budgets (128–512 tokens) exhibit faster decay, as expected, but do not collapse
 455 catastrophically.

456 C Implementation details

457 This appendix provides the full complexity derivation for fused top- κ selection, the pipelining
 458 schedule that masks Leiden latency, and the memory accounting for per-head graphs.

459 C.1 Fused top- κ selection

460 We construct the initial sparse graph by identifying the top- κ keys for all n queries simultaneously,
 461 fusing this selection directly into the FlashAttention kernel. As the kernel computes scores for each
 462 $b_q \times b_k$ tile of queries and keys, we apply a register-level bitonic sort [3] to identify local candidates.
 463 Block-level candidates are accumulated in query-specific buffers and reduced after processing the
 464 full history. The total complexity aggregates two costs across the causal mask:

- 465 1. **Block-level sorting.** We execute a bitonic sort for every tile in the causal mask. Since
 466 there are approximately $\frac{n}{b_q} \cdot \frac{n}{b_k} / 2$ total tiles, the aggregate block-level sorting complexity is
 467 $\mathcal{O}\left(\frac{n^2}{b_q b_k} \cdot b_k \log^2 b_k\right) = \mathcal{O}\left(\frac{n^2}{b_q} \log^2 b_k\right)$.
- 468 2. **Global top- κ reduction.** For query q_i , the candidate buffer accumulates approximately
 469 $\frac{i}{b_k} \kappa$ entries, assuming i tokens are partitioned into $\frac{i}{b_k}$ causal tiles that each contribute κ
 470 candidates. We employ radix select [18] to reduce this buffer and isolate the final sparse set
 471 $\mathcal{K}_i$. Summing this linear-time operation over n queries yields $\sum_{i=1}^n \mathcal{O}\left(\frac{i}{b_k} \kappa\right) = \mathcal{O}\left(\frac{n^2}{b_k} \kappa\right)$.

472 Combining these terms, the total graph construction complexity is:

$$\mathcal{O}\left(n^2 \left(\frac{\log^2 b_k}{b_q} + \frac{\kappa}{b_k}\right)\right). \quad (8)$$

473 Because the bracketed term depends solely on the fixed constants b_q , b_k , and κ , top- κ selection scales
 474 as $\mathcal{O}(n^2)$, matching the underlying matmul and ensuring that graph construction incurs no asymptotic
 475 overhead over standard prefill.

476 The empirical overhead aligns with this bound. With $b_k = 64$ or 128 and $\kappa = 8$, the theoretical
 477 overhead ratio κ/b_k falls between 6.25% and 12.5%. Wall-clock measurements on 130k-token
 478 inputs show prefill overhead of +7.8% (Qwen3-4B), +7.7% (Qwen3-8B), and +7.2% (Qwen3-14B),
 479 confirming the kernel adds only marginal cost over standard prefill (Table 5).

480 C.2 Pipelined graph partitioning

481 Naively, Leiden partitioning would extend the prefill critical path by the sum of per-layer partition
 482 costs. We instead dispatch each layer’s partitioning to asynchronous workers immediately after its
 483 attention scores are computed, allowing it to run in parallel with the FFN of the same layer and the
 484 attention of subsequent layers (Figure 4). Because the FFN and following layers’ attention together
 485 exceed the per-layer Leiden cost, all layers except the last have their partition cost fully masked.

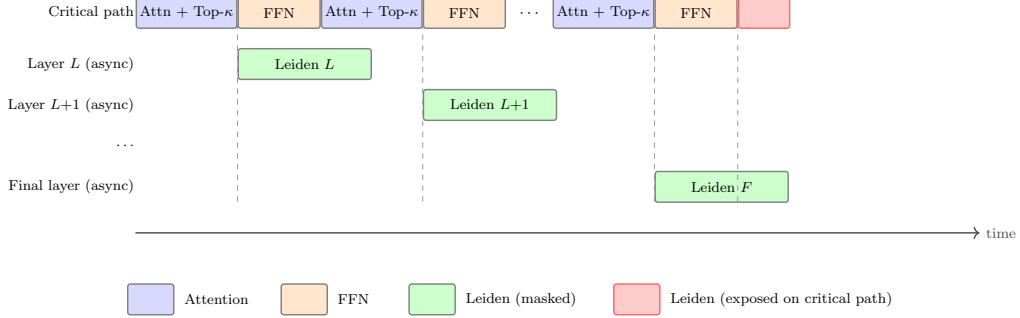


Figure 4: Pipeline schedule for a single decoding step. Leiden partitioning for each layer is dispatched asynchronously after its attention scores are computed, overlapping with subsequent layer execution. The red segment indicates the residual Leiden latency that remains exposed on the critical path when the final layer’s partitioning has not completed by the time it is needed.

486 The exposed latency is therefore the cost of partitioning a single layer’s graphs, which we measure at
 487 0.37–0.45 s across the Qwen3 family on an NVIDIA H100 (Table 5). This corresponds to 2–3% of
 488 total prefill wall time on a 130k-token input.

489 C.3 Memory footprint

490 CommunityKV maintains an independent attention graph per query head (or per KV group or per
 491 layer under aggregation). For modern multi-layer, multi-head models, the aggregate footprint of these
 492 graphs warrants explicit accounting.

493 **Per-graph footprint** Each graph stores at most $|E| \leq 2n\kappa(1 + \kappa)$ edges (Section 3.1). Storing
 494 edge weights in bfloat16 with 32-bit indices yields a per-graph cost of $8|E|$ bytes. At $n = 100k$
 495 tokens with $\kappa = 8$, this gives $|E| \approx 1.15M$ edges and a per-graph footprint of ~ 9 MB; the bound is
 496 < 20 MB at $\kappa = 8$ across all sequence lengths in our evaluation.

497 **Aggregate footprint** Without aggregation, a model with L layers and H query heads maintains
 498 $L \cdot H$ independent graphs. For Qwen3-14B ($L = 40$, $H = 40$, with 8 KV heads per layer), this is 1600
 499 per-head graphs. Under query-group aggregation (one graph per KV group, ablated in Section 4.4),
 500 this reduces to $L \cdot H_{KV} = 320$ graphs. At 100k tokens with $\kappa = 8$ in bfloat16, the aggregate footprint
 501 is < 15 GB without aggregation and < 3 GB with query-group aggregation, compared to ~ 30 GB for
 502 the dense KV cache at the same context length and the 28 GB Qwen3-14B model weights in bfloat16.
 503 Graph memory under query-group aggregation is therefore a small fraction of total model state.

504 **Centroid storage** Community centroids are stored in d -dimensional bfloat16. With $\mathcal{O}(\sqrt{n})$ com-
 505 munities per graph (Appendix A), centroid storage is $\mathcal{O}(\sqrt{n}d)$ per graph and is dominated by the
 506 edge storage above at all evaluated context lengths.

507 **Comparison to dense KV cache** The dense KV cache scales as $2L \cdot H_{KV} \cdot n \cdot d_{\text{head}}$ in bfloat16.
 508 For Qwen3-14B at 100k context, this is approximately 30 GB, an order of magnitude larger than
 509 the graph footprint under query-group aggregation. Because CommunityKV preserves the full KV
 510 cache (graphs serve as a routing index, not a replacement), graph memory is purely additive but small
 511 relative to the cache it indexes.

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